Assignment 2 — Linear Regression

SVNIT — Department of Artificial Intelligence

Question sheet: `Assignment-2.pdf`
Dataset: `../data/train.csv`

Plots are saved to `../plots/assignment_2/` and shown under each part.

In [1]:

```
import warnings
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r
2_score
from sklearn.model_selection import train_test_split

warnings.filterwarnings("ignore")
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")
pd.set_option("display.max_columns", 20)
pd.set_option("display.width", 120)


def find_project_root():
    starts = [Path.cwd().resolve()]
    for key in ("__vsc_ipynb_file__", "__session__"):
        val = globals().get(key)
        if val:
            starts.append(Path(val).resolve().parent)
    seen = set()
    for start in starts:
        for p in [start, *start.parents]:
            if p in seen:
                continue
            seen.add(p)
            if (p / "data" / "train.csv").exists():
                return p
    raise FileNotFoundError("Could not find data/train.csv.")


ROOT = find_project_root()
DATA = ROOT / "data"
PLOTS = ROOT / "plots" / "assignment_2"
PLOTS.mkdir(parents=True, exist_ok=True)


def section(title):
    print("\n" + "=" * 64)
    print(f"  {title}")
    print("=" * 64)


def pct(value, digits=2):
    return f"{100 * value:.{digits}f}%"


def money(value):
    return f"${value:,.2f}"
```

```python
def save_plot(name):
    path = PLOTS / name
    plt.tight_layout()
    plt.savefig(path, dpi=120, bbox_inches="tight")
    plt.close()
    print(f"Saved {path.relative_to(ROOT)}")


def print_metrics(y_true, y_pred, label):
    y_true = np.asarray(y_true)
    y_pred = np.asarray(y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    mae = mean_absolute_error(y_true, y_pred)
    r2 = r2_score(y_true, y_pred)
    mean_y = y_true.mean()
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100
    print(f"\n{label}")
    print(f"{'Metric':<28} {'Absolute':>16} {'As %':>12}")
    print("-" * 58)
    print(f"{'R2':<28} {r2:>16.4f} {pct(r2):>12}")
    print(f"{'MAE':<28} {money(mae):>16} {(mae / mean_y) * 100:>11.2
f}% of mean")
    print(f"{'RMSE':<28} {money(rmse):>16} {(rmse / mean_y) * 100:>11.
2f}% of mean")
    print(f"{'MSE':<28} {mse:>16,.0f}")
    print(f"{'MAPE':<28} {'':>16} {mape:>11.2f}%")
    return {"r2": r2, "mse": mse, "rmse": rmse, "mae": mae, "mape": ma
pe, "rmse_pct": (rmse / mean_y) * 100}


print("Project root:", ROOT)
print("Dataset:", DATA / "train.csv")
```

```
Project root: /Users/ojas/Desktop/ML Lab
Dataset: /Users/ojas/Desktop/ML Lab/data/train.csv
```

Part A: Exploratory Data Analysis

In [2]:
```
section("Part A: Exploratory Data Analysis")

df = pd.read_csv(DATA / "train.csv")
target = "SalePrice"
features = ["GrLivArea", "OverallQual", "GarageCars"]
model_df = df[[target] + features].dropna()

print(f"Rows used for modeling: {len(model_df):,}")
print(f"Target: {target}")
print(f"Predictors: {features}")

skew = model_df[target].skew()
print(f"\nTarget skewness: {skew:.3f}  ->  {'right-skewed' if skew >
0.5 else 'approximately symmetric'}")
print(f"Target mean:     {money(model_df[target].mean())}")
print(f"Target median:   {money(model_df[target].median())}")

plt.figure(figsize=(8, 5))
sns.histplot(model_df[target], kde=True, color="steelblue")
plt.axvline(model_df[target].mean(), color="red", ls="--", label=f"Mea
n {money(model_df[target].mean())}")
plt.axvline(model_df[target].median(), color="orange", ls="--", label=
f"Median {money(model_df[target].median())}")
plt.xlabel("SalePrice")
plt.ylabel("Count")
plt.title("SalePrice Distribution")
plt.legend()
save_plot("target_distribution.png")

numeric_df = df.select_dtypes(include="number").drop(columns=["Id", "M
SSubClass"], errors="ignore")
corr_sale = numeric_df.corr()[target].drop(target)
pos_feats = corr_sale.nlargest(3)
neg_feats = corr_sale.nsmallest(3)

print("\nStrongest POSITIVE correlations with SalePrice:")
print(f"{'Feature':<16} {'r':>8}")
print("-" * 26)
for feat, r in pos_feats.items():
    print(f"{feat:<16} {r:>+8.3f}")

print("\nStrongest NEGATIVE correlations with SalePrice:")
print(f"{'Feature':<16} {'r':>8}")
print("-" * 26)
for feat, r in neg_feats.items():
    print(f"{feat:<16} {r:>+8.3f}")

heatmap_cols = list(pos_feats.index) + list(neg_feats.index) + [targe
t]
plt.figure(figsize=(9, 7))
sns.heatmap(numeric_df[heatmap_cols].corr(), annot=True, fmt=".2f", cm
ap="coolwarm", center=0)
```

```
plt.title("Correlation Heatmap (Positive + Negative Features)")
save_plot("correlation_heatmap.png")

print("\nModeling features vs SalePrice:")
for feat, r in model_df.corr()[target].drop(target).items():
    print(f"{feat:<14} {r:>+8.3f}")

fig, axes = plt.subplots(1, 3, figsize=(14, 4.5))
for ax, feat in zip(axes, features):
    if feat == "OverallQual":
        sns.boxplot(x=model_df[feat], y=model_df[target], ax=ax, color
="steelblue")
    else:
        sns.scatterplot(x=model_df[feat], y=model_df[target], ax=ax, a
lpha=0.45, color="steelblue")
    ax.set_title(f"{feat} vs SalePrice  (r = {model_df[feat].corr(mode
l_df[target]):+.3f})")
save_plot("feature_scatter.png")
```

```
============================================================
  Part A: Exploratory Data Analysis
============================================================
Rows used for modeling: 1,460
Target: SalePrice
Predictors: ['GrLivArea', 'OverallQual', 'GarageCars']

Target skewness: 1.883  ->  right-skewed
Target mean:     $180,921.20
Target median:   $163,000.00
Saved plots/assignment_2/target_distribution.png

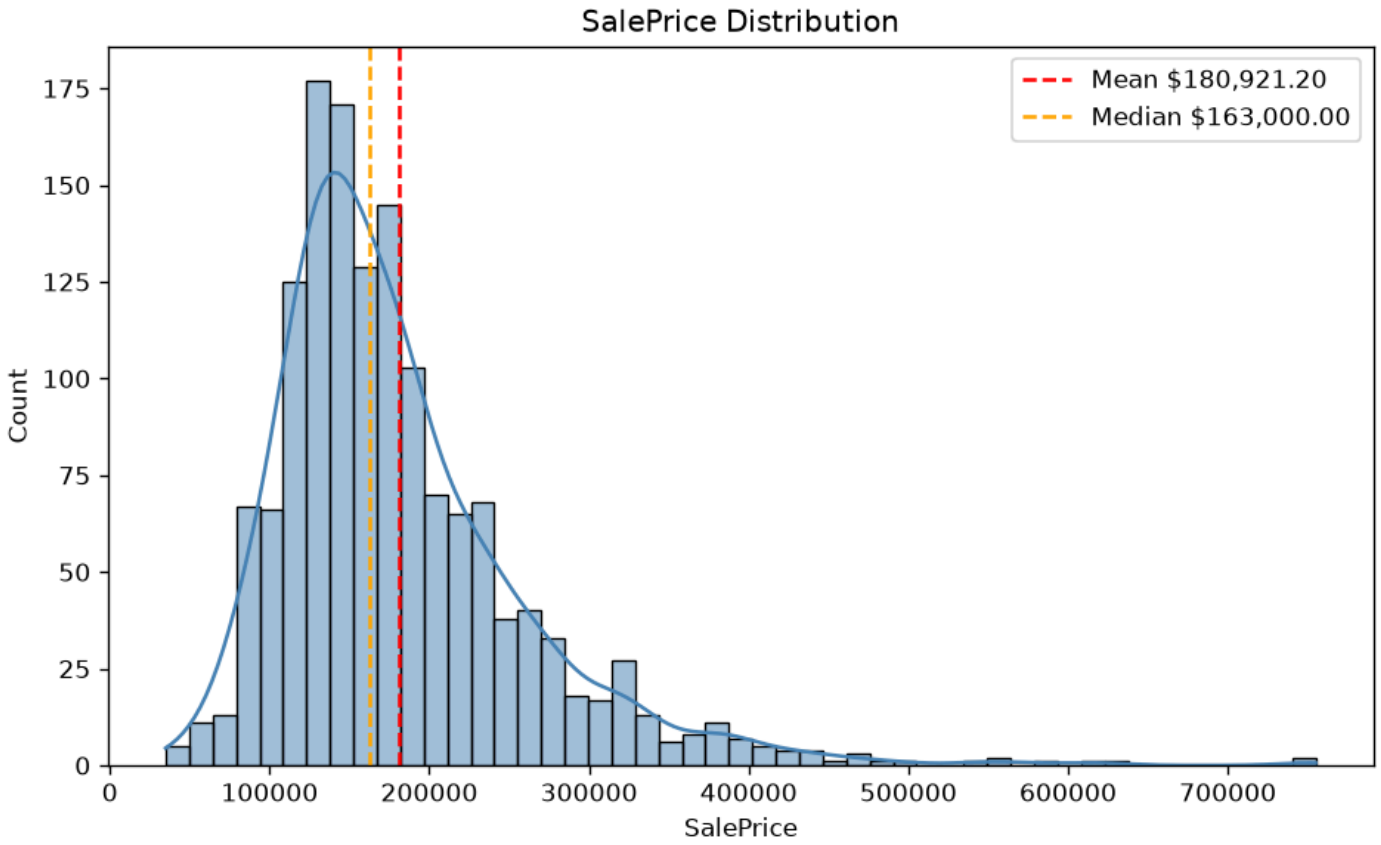
Strongest POSITIVE correlations with SalePrice:
Feature                 r
--------------------------
OverallQual        +0.791
GrLivArea          +0.709
GarageCars         +0.640

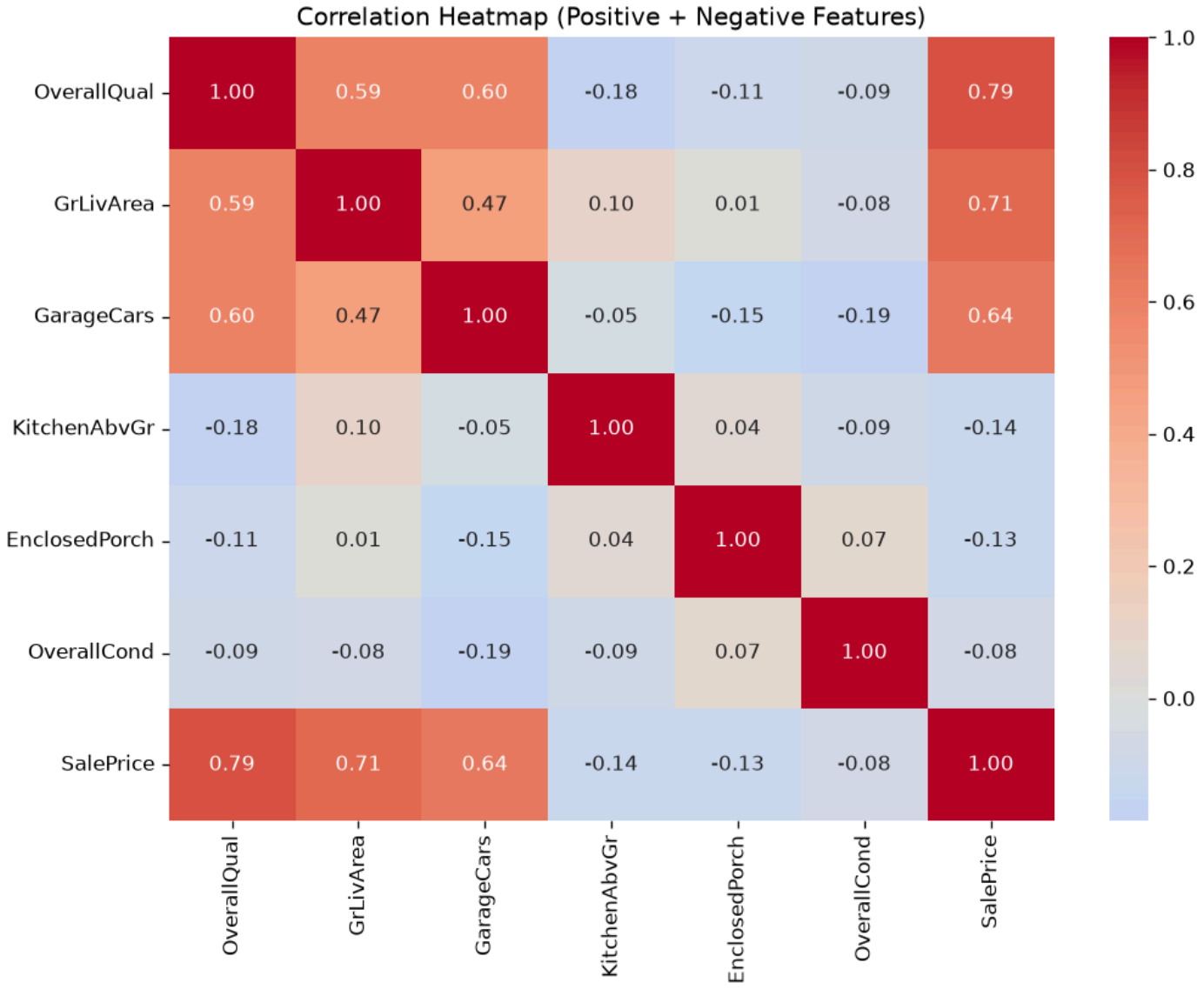
Strongest NEGATIVE correlations with SalePrice:
Feature                 r
--------------------------
KitchenAbvGr       -0.136
EnclosedPorch      -0.129
OverallCond        -0.078

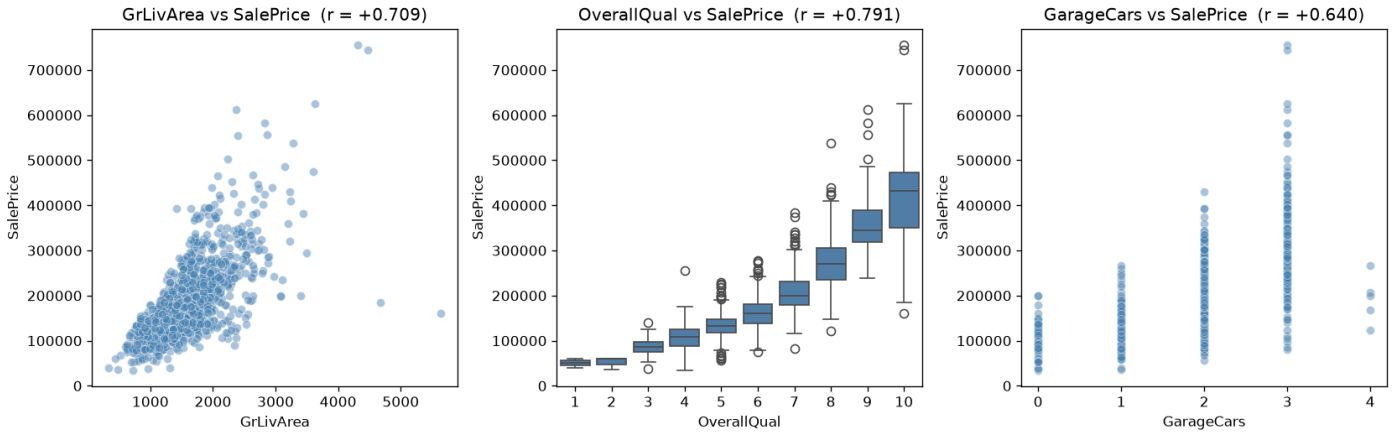
Saved plots/assignment_2/correlation_heatmap.png

Modeling features vs SalePrice:
GrLivArea        +0.709
OverallQual      +0.791
GarageCars       +0.640
Saved plots/assignment_2/feature_scatter.png
```

Part A graphs




Correlation Heatmap (Positive + Negative Features)
OverallQual
GrLivArea
GarageCars
KitchenAbvGr
EnclosedPorch
OverallCond
SalePrice
1.00 0.59 0.60 -0.18 -0.11 -0.09 0.79
0.59 1.00 0.47 0.10 0.01 -0.08 0.71
0.60 0.47 1.00 -0.05 -0.15 -0.19 0.64
-0.18 0.10 -0.05 1.00 0.04 -0.09 -0.14
-0.11 0.01 -0.15 0.04 1.00 0.07 -0.13
-0.09 -0.08 -0.19 -0.09 0.07 1.00 -0.08
0.79 0.71 0.64 -0.14 -0.13 -0.08 1.00
1.0
0.8
0.6
0.4
0.2
0.0


GrLivArea vs SalePrice (r = +0.709)
SalePrice
GrLivArea
OverallQual vs SalePrice (r = +0.791)
SalePrice
OverallQual
GarageCars vs SalePrice (r = +0.640)
SalePrice
GarageCars

Part B: Simple Linear Regression

In [3]:

```
section("Part B: Simple Linear Regression")

predictor = "GrLivArea"
X_train, X_test, y_train, y_test = train_test_split(
    model_df[[predictor]], model_df[target], test_size=0.2, random_sta
te=42
)

simple = LinearRegression().fit(X_train, y_train)
beta0, beta1 = simple.intercept_, simple.coef_[0]

print("Hypothesis:  y = b0 + b1 x + e")
print(f"Intercept b0: {beta0:,.2f}")
print(f"Slope     b1: {beta1:,.2f}")
print(f"Fitted: SalePrice = {beta0:,.2f} + {beta1:,.2f} * GrLivArea")
print(f"Interpretation: +1 sq ft in GrLivArea -> {money(beta1)} change
in SalePrice on average.")

plt.figure(figsize=(8, 6))
sns.scatterplot(x=model_df[predictor], y=model_df[target], alpha=0.5,
color="steelblue")
x_line = np.linspace(model_df[predictor].min(), model_df[predictor].ma
x(), 100)
y_line = simple.predict(pd.DataFrame({predictor: x_line}))
plt.plot(x_line, y_line, "r", lw=2, label="Regression line")
plt.xlabel(predictor)
plt.ylabel(target)
plt.title("Simple Linear Regression: GrLivArea vs SalePrice")
plt.legend()
save_plot("simple_regression.png")

y_pred_s = simple.predict(X_test)
simple_metrics = print_metrics(y_test, y_pred_s, "Simple Linear Regres
sion (test)")

fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.scatterplot(x=y_test, y=y_pred_s, ax=axes[0], alpha=0.6, color="st
eelblue")
lo = min(y_test.min(), y_pred_s.min())
hi = max(y_test.max(), y_pred_s.max())
axes[0].plot([lo, hi], [lo, hi], "r--", label="Perfect prediction")
axes[0].set_xlabel("Actual SalePrice")
axes[0].set_ylabel("Predicted SalePrice")
axes[0].set_title("Actual vs Predicted (simple)")
axes[0].legend()

resid_s = y_test - y_pred_s
sns.scatterplot(x=y_pred_s, y=resid_s, ax=axes[1], alpha=0.6, color="s
teelblue")
axes[1].axhline(0, color="r", ls="--")
axes[1].set_xlabel("Predicted SalePrice")
axes[1].set_ylabel("Residuals")
```

```
axes[1].set_title("Residual Plot (simple)")
save_plot("simple_diagnostics.png")
```

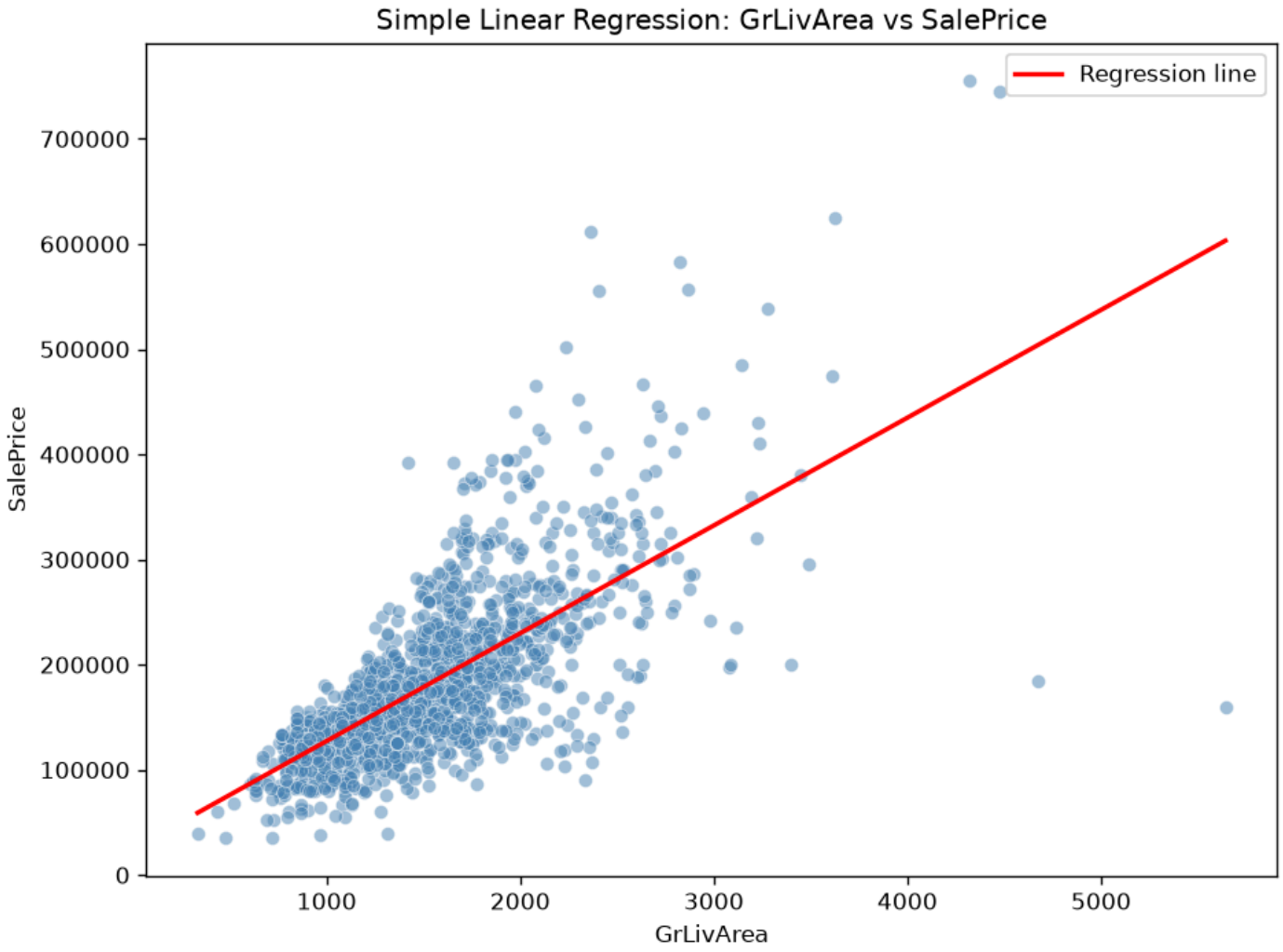
```
============================================================
  Part B: Simple Linear Regression
============================================================
Hypothesis:  y = b0 + b1 x + e
Intercept b0: 24,899.75
Slope     b1: 102.49
Fitted: SalePrice = 24,899.75 + 102.49 * GrLivArea
Interpretation: +1 sq ft in GrLivArea -> $102.49 change in SalePrice o
n average.

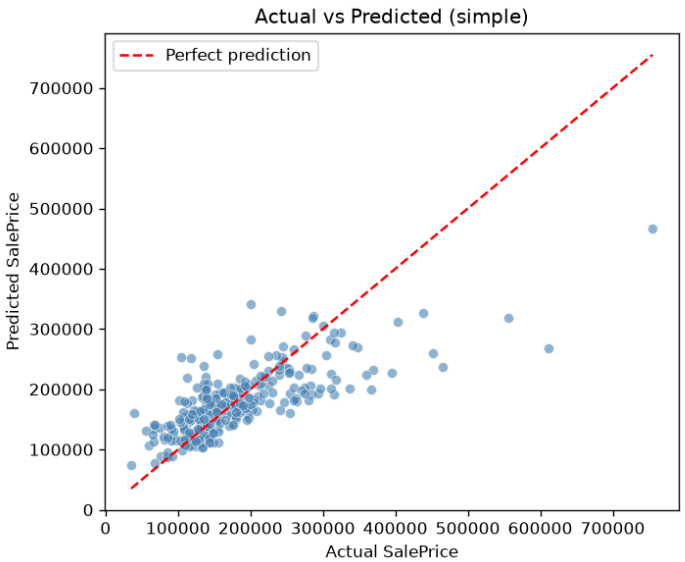
Saved plots/assignment_2/simple_regression.png

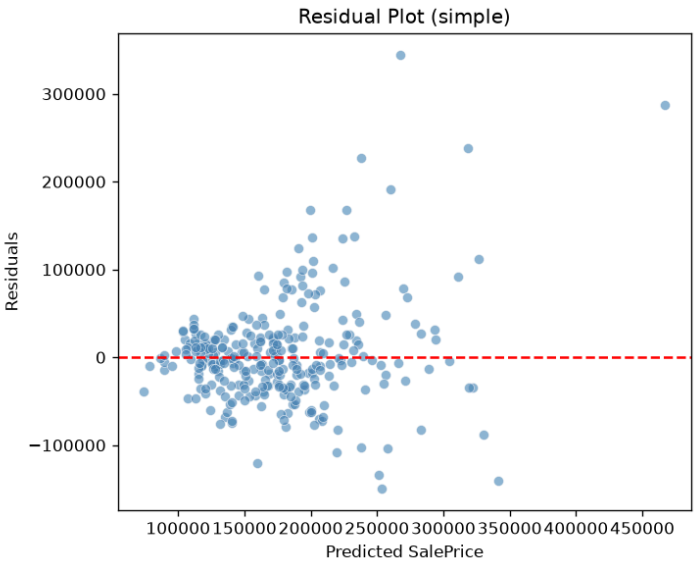
Simple Linear Regression (test)
Metric                                  Absolute         As %
------------------------------------------------------------
R2                                        0.5543       55.43%
MAE                                     $38,341.20       21.44% of mean
RMSE                                    $58,471.76       32.70% of mean
MSE                                  3,418,946,311
MAPE                                                     23.34%

Saved plots/assignment_2/simple_diagnostics.png
```

Part B graphs







Part C: Multiple Linear Regression

In [4]:

```
section("Part C: Multiple Linear Regression")

X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(
    model_df[features], model_df[target], test_size=0.2, random_state=
42
)
multi = LinearRegression().fit(X_train_m, y_train_m)
y_pred_m = multi.predict(X_test_m)

print("Fitted equation:")
eq = f"  SalePrice = {multi.intercept_:,.2f}"
for f, c in zip(features, multi.coef_):
    eq += f" {'+' if c >= 0 else '-'} {abs(c):,.2f}*{f}"
print(eq)

print("\nCoefficients:")
print(f"{'Feature':<14} {'Coefficient':>14}")
print("-" * 30)
print(f"{'Intercept':<14} {multi.intercept_:>14,.2f}")
for f, c in zip(features, multi.coef_):
    print(f"{f:<14} {c:>14,.2f}")

print(f"\nInterpretation (OverallQual): +1 quality point -> {money(mul
ti.coef_[1])} SalePrice, other predictors held fixed.")

multi_metrics = print_metrics(y_test_m, y_pred_m, "Multiple Linear Reg
ression (test)")

print("\nModel comparison")
print(f"{'Model':<22} {'R2 %':>8} {'RMSE %':>10} {'MAPE %':>10} {'RMS
E':>14}")
print("-" * 68)
print(
    f"{'Simple (GrLivArea)':<22} {pct(simple_metrics['r2']):>8} "
    f"{simple_metrics['rmse_pct']:>9.2f}% {simple_metrics['mape']:>9.2
f}% {money(simple_metrics['rmse']):>14}"
)
print(
    f"{'Multiple (3 feats)':<22} {pct(multi_metrics['r2']):>8} "
    f"{multi_metrics['rmse_pct']:>9.2f}% {multi_metrics['mape']:>9.2
f}% {money(multi_metrics['rmse']):>14}"
)

print("\nWinner: Multiple regression")
print(f"  Extra variance explained: {pct(multi_metrics['r2'] - simple_
metrics['r2'])}")
print("  Why: quality and garage size add information beyond living ar
ea.")

residuals = y_test_m - y_pred_m
lo = min(y_test_m.min(), y_pred_m.min())
hi = max(y_test_m.max(), y_pred_m.max())
```

```
plt.figure(figsize=(8, 5))
sns.scatterplot(x=y_pred_m, y=residuals, alpha=0.6, color="steelblue")
plt.axhline(0, color="r", ls="--")
plt.xlabel("Predicted SalePrice")
plt.ylabel("Residuals")
plt.title("Residual Plot — Multiple Linear Regression")
save_plot("residual_plot.png")

plt.figure(figsize=(7, 7))
sns.scatterplot(x=y_test_m, y=y_pred_m, alpha=0.6, color="steelblue")
plt.plot([lo, hi], [lo, hi], "r--", label="Perfect prediction")
plt.xlabel("Actual SalePrice")
plt.ylabel("Predicted SalePrice")
plt.title("Actual vs Predicted — Multiple Linear Regression")
plt.legend()
save_plot("actual_vs_predicted.png")

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))
names = ["Simple\n(GrLivArea)", "Multiple\n(3 features)"]
colors = ["steelblue", "seagreen"]
axes[0].bar(names, [simple_metrics["r2"], multi_metrics["r2"]], color=
colors)
axes[0].set_ylim(0, 1)
axes[0].set_ylabel("R²")
axes[0].set_title("R² (higher is better)")
for i, v in enumerate([simple_metrics["r2"], multi_metrics["r2"]]):
    axes[0].text(i, v + 0.02, f"{v:.2%}", ha="center")
axes[1].bar(names, [simple_metrics["rmse"], multi_metrics["rmse"]], co
lor=colors)
axes[1].set_ylabel("RMSE ($)")
axes[1].set_title("RMSE (lower is better)")
for i, v in enumerate([simple_metrics["rmse"], multi_metrics["rms
e"]]):
    axes[1].text(i, v + 800, f"${v:,.0f}", ha="center")
save_plot("model_comparison.png")
```

```
============================================================
  Part C: Multiple Linear Regression
============================================================
Fitted equation:
  SalePrice = -94,208.72 + 47.23*GrLivArea + 26,772.49*OverallQual + 2
2,239.46*GarageCars

Coefficients:
Feature           Coefficient
------------------------------
Intercept          -94,208.72
GrLivArea               47.23
OverallQual         26,772.49
GarageCars          22,239.46

Interpretation (OverallQual): +1 quality point -> $26,772.49 SalePric
e, other predictors held fixed.

Multiple Linear Regression (test)
Metric                                  Absolute         As %
-------------------------------------------------------------
R2                                        0.7586       75.86%
MAE                                   $28,121.52       15.72% of mean
RMSE                                  $43,032.15       24.06% of mean
MSE                                1,851,765,975
MAPE                                                    17.55%

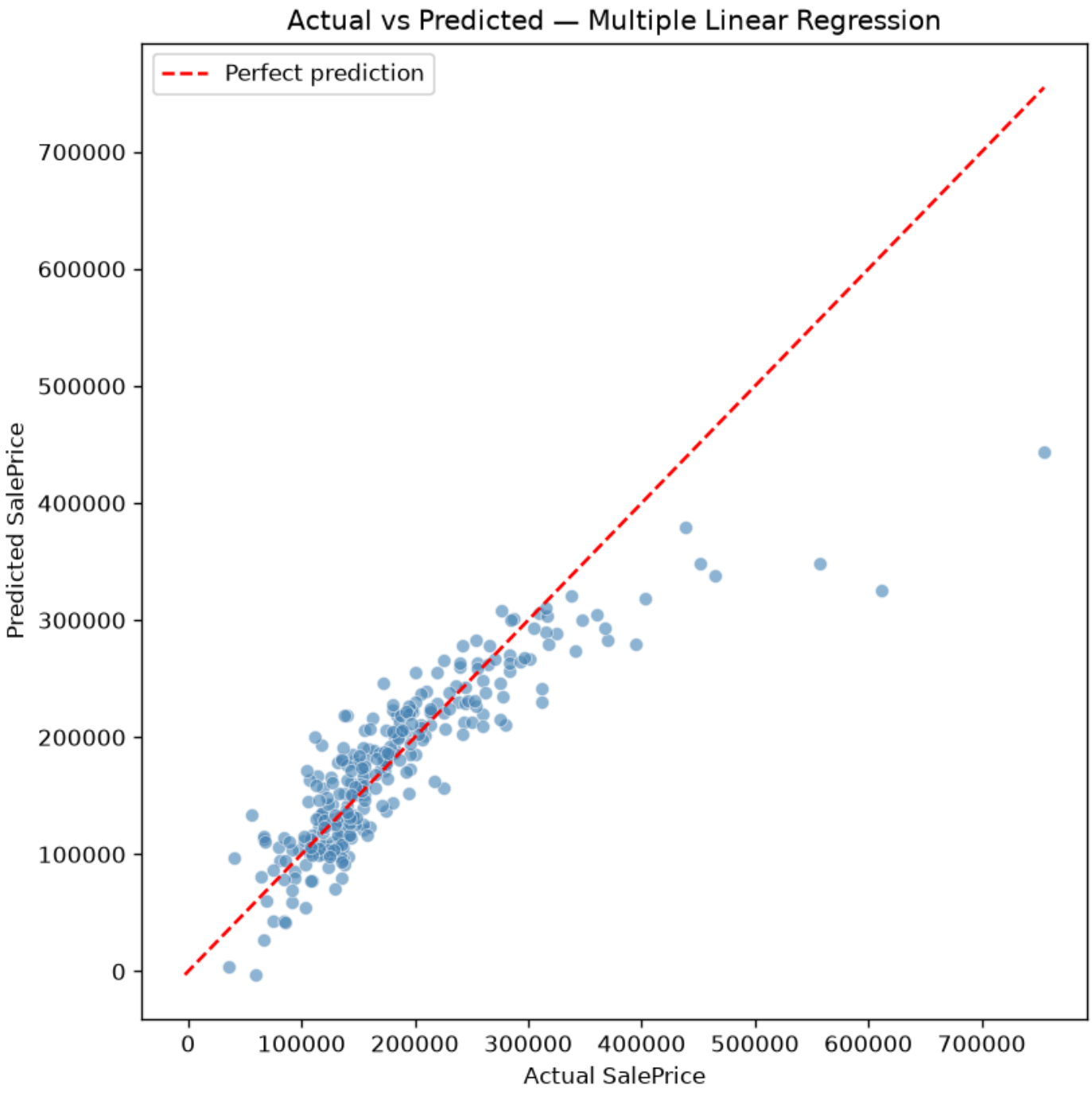
Model comparison
Model                      R2 %     RMSE %     MAPE %           RMSE
-----------------------------------------------------------------------
Simple (GrLivArea)       55.43%     32.70%     23.34%     $58,471.76
Multiple (3 feats)       75.86%     24.06%     17.55%     $43,032.15

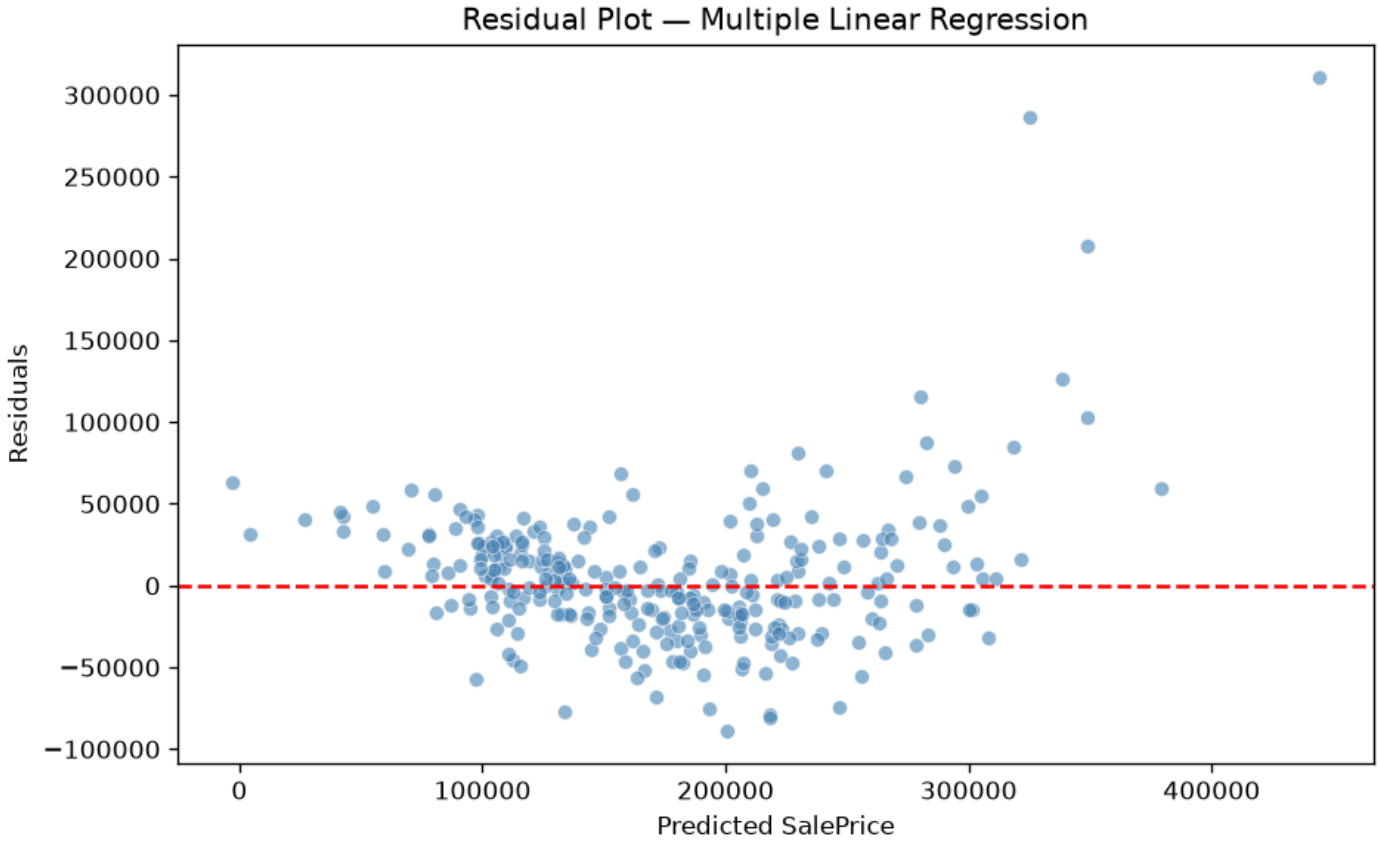
Winner: Multiple regression
  Extra variance explained: 20.43%
  Why: quality and garage size add information beyond living area.

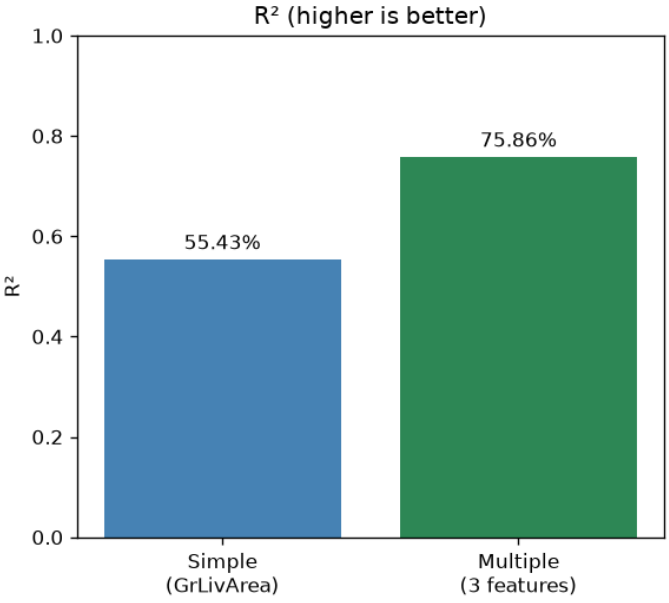
Saved plots/assignment_2/residual_plot.png
Saved plots/assignment_2/actual_vs_predicted.png

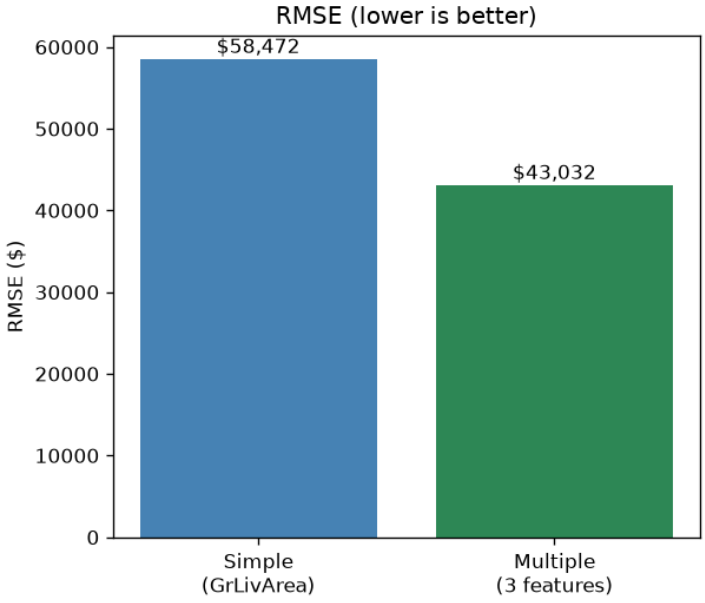
Saved plots/assignment_2/model_comparison.png
```

Part C graphs




Residual Plot — Multiple Linear Regression
Residuals
Predicted SalePrice
300000
250000
200000
150000
100000
50000
0
−50000
−100000
0
100000
200000
300000
400000


R² (higher is better)
R²
1.0
0.8
0.6
0.4
0.2
0.0
55.43%
75.86%
Simple
(GrLivArea)
Multiple
(3 features)


RMSE (lower is better)
RMSE ($)
60000
50000
40000
30000
20000
10000
0
$58,472
$43,032
Simple
(GrLivArea)
Multiple
(3 features)